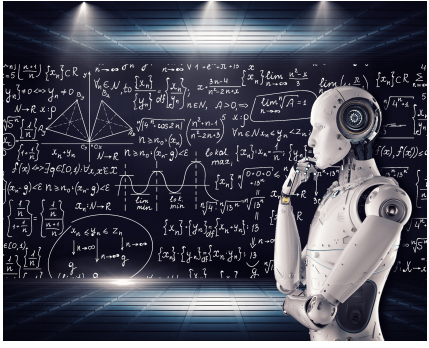
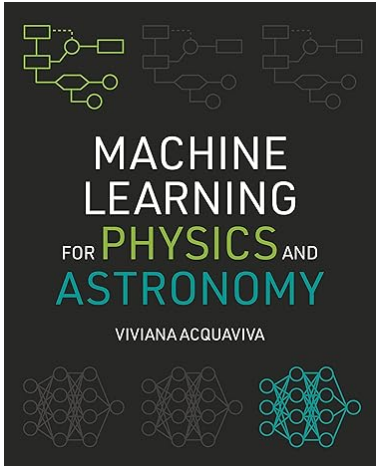
Introduction to Machine Learning Methods

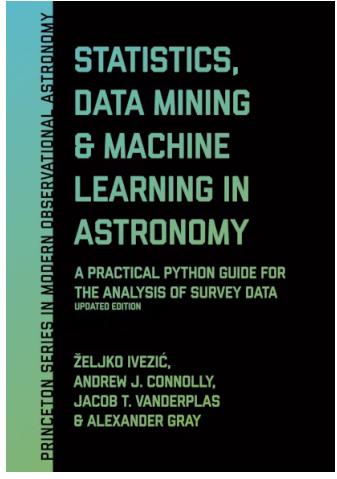
Lecture/Lab 21

Physics 4730, Fall 2026



References


MACHINE
LEARNING
FOR PHYSICS AND
ASTRONOMY
VIVIANA ACQUAVIVA


PRINCETON SERIES IN MODERN OBSERVATIONAL ASTRONOMY
STATISTICS, DATA MINING & MACHINE LEARNING IN ASTRONOMY
A PRACTICAL PYTHON GUIDE FOR THE ANALYSIS OF SURVEY DATA
UPDATED EDITION
ŽELJKO IVEZIĆ,
ANDREW J. CONNOLLY,
JACOB T. VANDERPLAS
& ALEXANDER GRAY

What is Machine Learning?

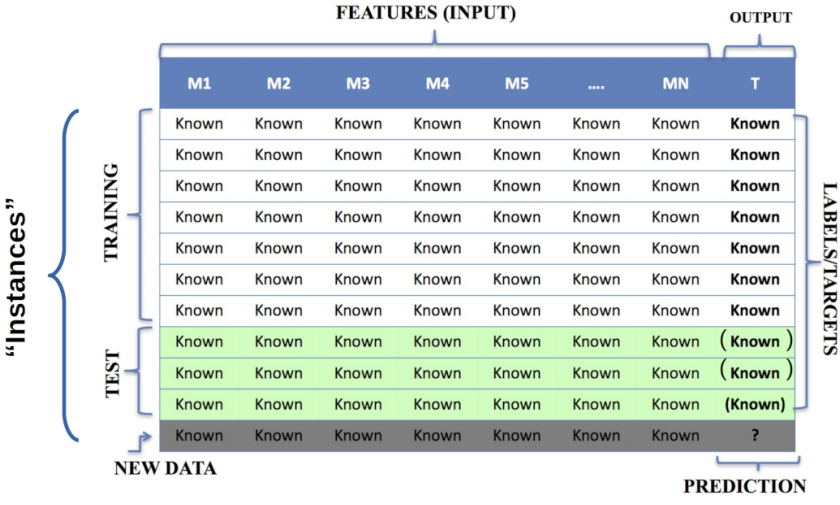
- The process of teaching a machine to make informed, data-driven decisions.

What Can We Do With It?

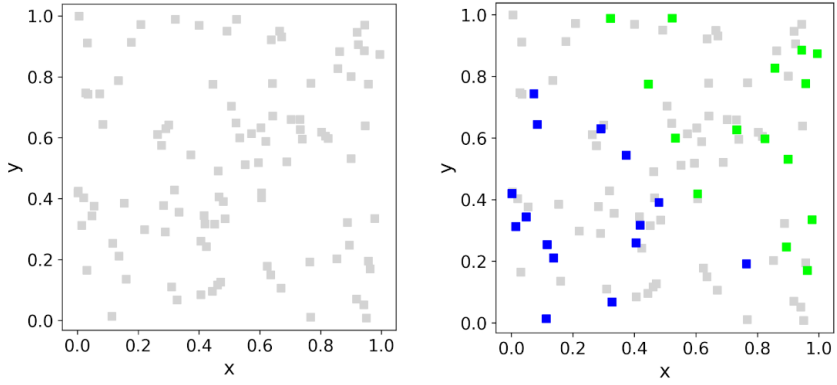


- **Recognize** an instance of a certain type. *e.g.* identify an animal as a cat.
- **Predict** some property or information on the basis of other information. *e.g.* predict whether my cat will like some food, on the basis of other food she likes.
- **Group** similar objects. *e.g.* separate long-haired cats from short-haired cats in a collection of cats.
- **Simplify** information to it's essential nature. *e.g.* make the simplest drawing of a cat that is still recognizable as a cat.

The Language of Machine Learning



Supervised (vs Unsupervised) Learning



Supervised learning problem: Learn from the **training set** to identify squares in the larger set as blue or green.

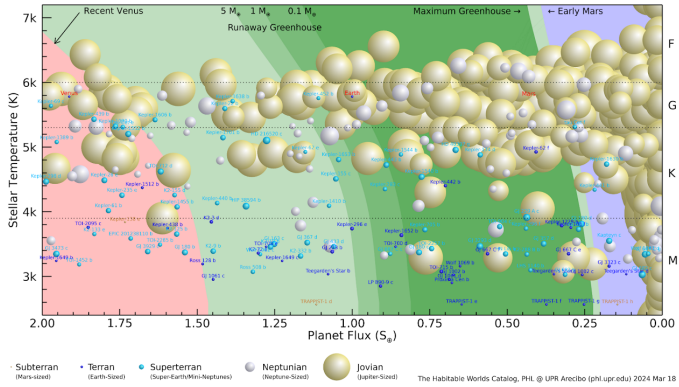
Building an ML Model

Fundamental Steps:

- Arrange the data into features and output arrays
- Split them into a training set and a test set
- Select the ML algorithm you want to use, and its parameters
- Apply the selected algorithm to the training set, which will create a tentative features-output relationship.
- Apply the model to the test data to predict the target property
- Evaluate the model by comparing predicted/true test target property.
- Rejoice, or repeat.

Building an ML Model: The Search for Habitable Planets

- **phl.upr.edu/hwc**
- Subset of 18 instances
- Three features:
  - Mass of the star
  - Orbital period of planet
  - Distance between planet and star



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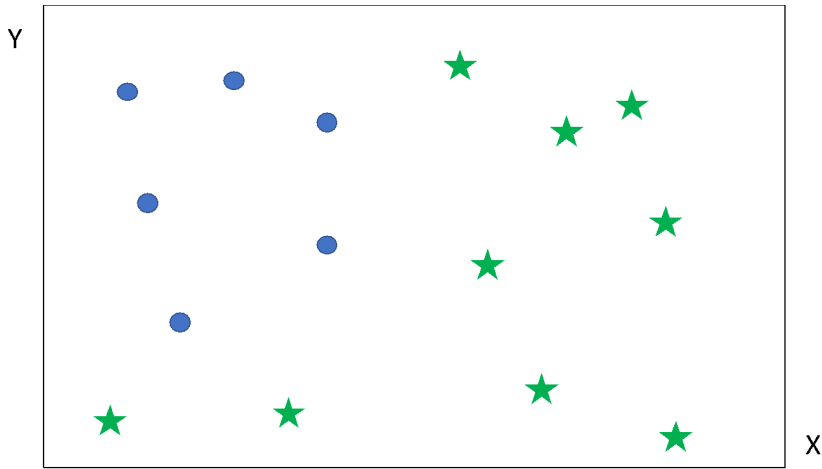
Name	Stellar Mass ($M_\odot$)	Orbital Period (days)	Distance (AU)	Habitable?
Kepler-736 b	0.86	3.60	0.0437	0
Kepler-636 b	0.85	16.08	0.1180	0
Kepler-887 c	1.19	7.64	0.0804	0
Kepler-442 b	0.61	112.30	0.4093	1
Kepler-772 b	0.98	12.99	0.1074	0
Teegarden's Star b	0.09	4.91	0.0252	1
K2-116 b	0.69	4.66	0.0481	0
GJ 1061 c	0.12	6.69	0.035	1
HD 68402 b	1.12	1103	2.1810	0
Kepler-1544 b	0.81	168.81	0.5571	1
Kepler-296 e	0.5	34.14	0.1782	1
Kepler-705 b	0.53	56.06	0.2319	1
Kepler-445 c	0.18	4.87	0.0317	0
HD 104067 b	0.62	55.81	0.26	0
GJ 4276 b	0.41	13.35	0.0876	0
Kepler-296 f	0.5	63.34	0.2689	1
Kepler-63 b	0.98	9.43	0.0881	0
GJ 3293 d	0.42	48.13	0.1953	1

Building an ML Model: The Search for Habitable Planets

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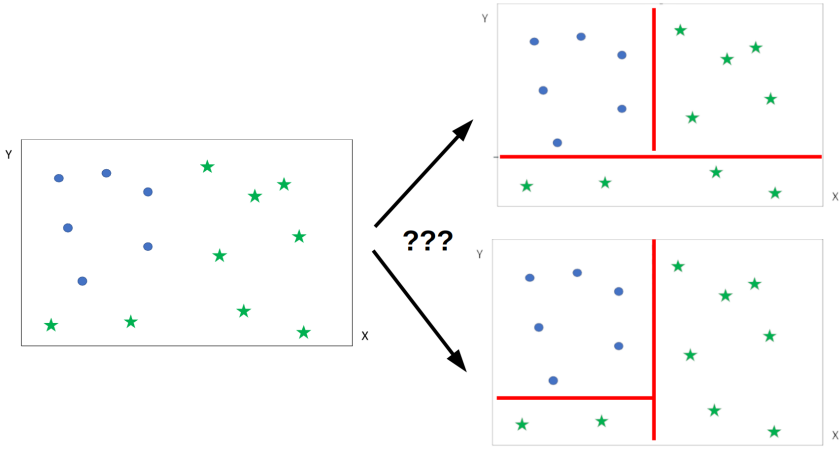
	Name	Stellar Mass ($M_\odot$)	Orbital Period (days)	Distance (AU)	Habitable?
Training	Kepler-736 b	0.86	3.60	0.0437	0
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	Kepler-445 c	0.18	4.87	0.0317	0
Test	HD 104067 b	0.62	55.81	0.26	
	GJ 4276 b	0.41	13.35	0.0876	
	Kepler-296 f	0.5	63.34	0.2689	
	Kepler-63 b	0.98	9.43	0.0881	
	GJ 3293 d	0.42	48.13	0.1953	

Decision Trees



How might you sort the above data?

Decision Trees



Is one option better than the other?

Decision Trees: Gini Impurity

$$G = 1 - \sum_{i=1}^{N_{\text{class}}} f(i)^2$$

where $f(i)$ is the fractional abundance of each class.

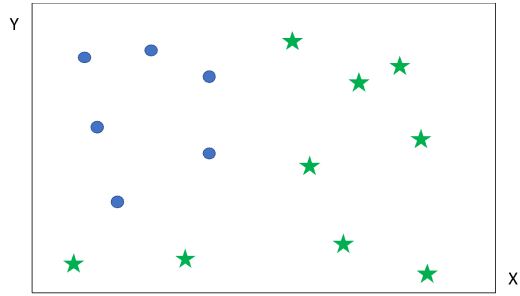


Corrado Gini (1884-1965)

Decision Trees: Gini Impurity

Original Data:

$$
\begin{aligned}
G_{orig} &= 1 - \sum_{i=1}^{N_{\text{class}}} f(i)^2 \\
&= 1 - \left(\frac{6}{15}\right)^2 - \left(\frac{9}{15}\right)^2 \\
&= 0.480
\end{aligned}
$$

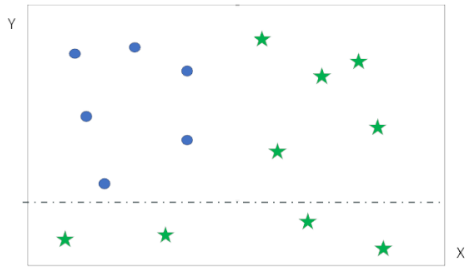


Decision Trees: Gini Impurity

Horizontal Split First:

$$G_{lower} = 1 - \left(\frac{4}{4}\right)^2 - \left(\frac{0}{4}\right)^2 = 0.000$$

$$G_{upper} = 1 - \left(\frac{6}{11}\right)^2 - \left(\frac{5}{11}\right)^2 = 0.496$$



Total decrease in impurity

$$\Delta G = 0.480 - \left(\frac{4}{15} \times 0\right) - \left(\frac{11}{15} \times 0.496\right)$$

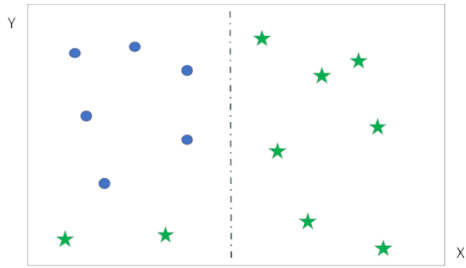
$$= 0.116$$

Decision Trees: Gini Impurity

Vertical Split First:

$$G_{left} = 1 - \left(\frac{2}{8}\right)^2 - \left(\frac{6}{8}\right)^2 = 0.375$$

$$G_{right} = 1 - \left(\frac{7}{7}\right)^2 - \left(\frac{0}{7}\right)^2 = 0.00$$



Total decrease in impurity

$$\Delta G = 0.480 - \left(\frac{8}{15} \times 0.375\right) - \left(\frac{7}{15} \times 0\right)$$

$$= 0.280 \longrightarrow \text{this is the preferred option!}$$

Lab 21

- Use **Gini Impurity** as a way to separate data with different target output. We'll work in 1-dimension to make things simple.
- This will give us a feel for how **scikit-learn** uses Gini Impurity in forming decision trees.
- Lab 22 (next time) we'll start employing **scikit-learn** tools.
- p.s. I owe you a homework assignment...

